

Md Tahmidur Rafid

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Education

- 2017 – 2022  **B.Sc. in CSE, Bangladesh University of Engineering and Technology (BUET)**
CGPA: 3.83
Thesis title: *An approach to generate scale-free outerplanar networks.*
Research Area: Graph theory, Network science, Algorithms and theory.

Research Publications

Conference Proceedings

- 1 M. T. Rafid, R. Ibrat, and M. S. Rahman, “Generating scale-free outerplanar networks,” in *New Trends in Computer Technologies and Applications*, Singapore: Springer Nature Singapore, 2022, pp. 156–166.
Link: https://link.springer.com/chapter/10.1007/978-981-19-9582-8_14
Role: First Author.
Conference: International Computer Symposium (ICS 2022).

Research Experience

- 2021 – 2022  **Characterizing the degree sequence of maximal outerplanar graphs.**
- Supervisor: Dr. Md. Saidur Rahman.
- Our initial motive was to find the complete characterization of the degree sequences of maximal outerplanar graphs, but it's still an open problem. We did find some rules for certain cases. We also came up with an $O(n)$ time algorithm that creates this type of graphs from a sequence of degrees following certain criteria.
-  **Generating scale-free planar networks..**
- Supervisor: Dr. Md. Saidur Rahman.
- Usually, the scale-free property conflicts planarity. Thus, we found that topic interesting to combine those two attributes. We came up with a theorem regarding the existence of scale-free maximal outerplanar networks. Additionally, we've developed an algorithm to form networks those are both scale-free and outerplanar.

Research Interests

-  I have initial research experience in the fields of graph theory, network science, algorithms and theory. However, I am highly interested in exploring other areas such as software engineering, machine learning, computer security, computer networks, HCI, and so on.

Professional Experience

- 2022 – Present  **Software Developer, IQVIA.**
- Designed and managed **microservice** architecture with support for multiple databases and **multitenancy**.
- Collaborated with DevOps for **CI/CD** processes and version control with Git.
- Developed and utilized **in-house libraries** across different projects.
- Experienced in developing applications using .NET Core, EF Core, Identity Server, SQL, NoSQL, redis, rabbitMQ, Angular etc.

Miscellaneous Experience

Awards and Achievements

- 2021  **Best Paper**, 2nd Best Paper Award at IEEE Computer Society Bangladesh Chapter Summer Symposium 2021.
- 2017-2022  **Dean's List Award**, Received Dean's list award and technical scholarships in each end of the academic years from the university.
- 2021  **ICT Competition**, Winner at Huawei ICT Competition (National) 2021.
- 2023  **Platinum Award**, IQVIA Impact program.
- 2020-2021  **Codeforces**, Max contest rating: 2089.

Skills

- Languages  C++, Python, Java, C#, Typescript, HTML, CSS.
- Frameworks  .NET Core, Angular, Spring, Express, Vue.js.
- Databases  MySQL, PostgreSQL, MSSQL, MongoDB.
- Technologies  Git, Docker, Unit Testing, BDD Testing, Contract Testing, CI/CD.

Projects

-  **Remote Whiteboard**
 - An app with **9k+ downloads** and **4.8+ rating** on Playstore.
 - Utilization of technologies such as WiFi network, browser-based UI, and touch screen support.
-  **Interactive Code Learning Platform**
 - Interactive learning platform with the motto of increasing user engagement apart from traditional learning.
 - Technology Stack: Vue.js, Node.js, Express(Framework), Mysql.
-  **Be A Donor**
 - A donation making platform.
 - Technology Stack: Spring Boot, JPA/Hibernate, Sitemesh, Bootstrap, MySQL.

References

Dr. Md. Saidur Rahman
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Md. Shariful Islam Bhuyan
Assistant Professor
CSE, BUET,
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